

Learning of Bash Scripting

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1. What is Bash?

Bash (short for **Bourne Again Shell**) is a command-line interpreter that provides a text-based interface for interacting with Unix and Linux operating systems. It allows users to execute commands, automate repetitive tasks, and combine multiple actions into executable scripts.

2. The Shebang (`#!/bin/bash`)

The very first line of a Bash script.

This line is known as a **shebang** (or hashbang). It tells the operating system's kernel to execute the rest of the script using the Bash interpreter located at `/bin/bash`. Without it, the system would default to whatever shell the current user is running, which might cause script compatibility issues.

3. Essential Commands

Command	Full Name	Description
<code>pwd</code>	Print Working Directory	Displays the absolute path of the current directory.
<code>ls</code>	List	Lists files and directories in the current folder.
<code>cd</code>	Change Directory	Navigates to a different directory.
<code>echo</code>	Echo	Prints a line of text or variable value to the terminal.
<code>touch</code>	Touch	Creates an empty file or updates the timestamp of an existing file.
<code>mkdir</code>	Make Directory	Creates a new folder.
<code>cat</code>	Concatenate	Displays the contents of a file or combines files.
<code>cp</code>	Copy	Copies files or directories from one location to another.
<code>mv</code>	Move	Moves or renames files and directories.
<code>rm</code>	Remove	Deletes files or folders (use <code>-r</code> for folders).